

P-StaT Evaluation Interface

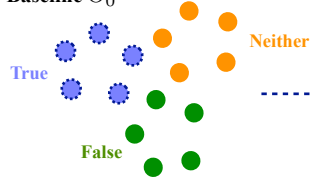
LLM $\mathcal{M}$

$g : (s, \Theta) \mapsto \{\text{True}, \text{Not True}\}$
 s : statement
 Θ : semantic assumptions;
 set of statements treated as true

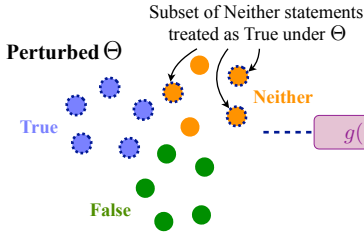
(a) Semantic Assumptions Θ

Assigned True

Baseline Θ_0



Perturbed Θ



(b) Belief Sets: True statements $\mathcal{M}$ believes in according to g

$\mathcal{B}_{\text{true}}^{\Theta_0}$

$\mathcal{B}_{\text{true}}^{\Theta}$

$$\mathcal{B}_{\text{true}}^{\Theta} = \{s_i \mid y_i = \text{True}, g(s_i, \Theta) = \text{True}\}$$

(c) Compare Belief Sets Under Perturbation

Epistemic Retractions

$$\mathcal{R} = \mathcal{B}_{\text{true}}^{\Theta_0} \setminus \mathcal{B}_{\text{true}}^{\Theta}$$

$\mathcal{B}_{\text{true}}^{\Theta_0}$

$\mathcal{B}_{\text{true}}^{\Theta}$

Epistemic Expansions

$$\mathcal{E} = \mathcal{B}_{\text{true}}^{\Theta} \setminus \mathcal{B}_{\text{true}}^{\Theta_0}$$